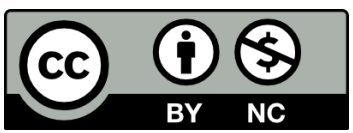
1 SOFTWARE ENGINEERING

course "software requirements and architecture"

Oct 2021

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Characterisation of the discipline

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definition

- Software engineering corresponds to the utilisation of the basic engineering principles to obtain, in a economically-viable way, reliable software that runs efficiently in real computers.
- Engineers must be able to manage the development and industrialisation of useful and efficient system.
- Software engineering is not just centred in the technical aspects associated with the development, but includes also activities related to managing the development process.
- Small software programs are not generally considered sufficiently complex to demand the use of an engineering approach.

definition

software engineering: application of a systematic, disciplined and quantifiable approach in the context of the planning, development and exploration of software systems.

- Software engineering is an engineering discipline that is focused on all aspects concerning the development of software-based systems.
- Succinctly, it is the application of engineering to the software domain.

SWEBOK knowledge areas

SWEBOK (software engineering body of knowledge): reference to characterise the software engineering discipline.

1	software requirements	9	software eng. models and methods
2	software design	10	software quality
3	software construction	11	software eng. professional practice
4	software testing	12	software engineering economics
5	software maintenance	13	computing foundations
6	softw. configuration manag.	14	mathematics foundations
7	softw. engineering management	15	engineering foundations
8	software engineering process		

SWEBOK | KA1: software requirements

- This KA handles the elicitation, analysis, documentation, validation, and maintenance of software requirements.
- Requirements are considered as properties that the systems (still in project) may manifest later after development.
- The software requirements express the necessities and the restrictions that are put to a software system.

SWEBOK | KA2: software design



- Software design (KA2) is the process where the architecture, components, interfaces, and other system characteristics are defined.
- It is the activity in which the software requirements are handled for describing the internal structure of the system.
- The final result of the process should describe:
  1. the system architecture,
  2. the interfaces between the components,
  3. the components with a level of detail that permits their construction.

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main characteristics

- Software engineering concerns the development of large/complex applications.
- The fundamental issue is mastering complexity.
- Software evolves.
- The efficiency with which software is developed is of crucial importance.
- Regular cooperation between people occurs within the development of a software applications.
- The software has to support its users effectively.
- Software engineering is a balancing act.

[van Vliet, 2007]

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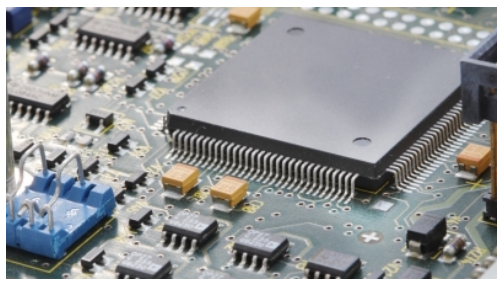
Software systems and products

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definition

- The term ‘software’ was used for the first time in 1959.
- Applied Data Research, founded in 1959, was the 1st company to sell software products separated from the hardware.
- In 1970, the American justice demanded IBM to distinguish, in its accounting documents, hardware and software, providing separate prices for each one.
- One can define software by a process of elimination.
- Everything that, in the users perspective, composes a computer, except the hardware, is the software [Galler, 1962].
- The hardware of a computer is composed of electronic and electromechanical components (CPU, memory, I/O devices).





definition

Software:

- the programs, which include the instructions that are executed by the computer (more specifically the hardware), as well as the data that are operated by those instructions.
- the algorithms (instructions that describe in detail how to accomplish something) and their respective representations (the programs).
- the set of programs, procedures and rules (and occasionally documentation), related to the operation of a system that aims to acquire, store, and process the data.
- the abstract element that, together with the hardware, constitutes the automatised part of a real-world system, implementing a stimulus-answer mechanism, with the objective of satisfying the needs of the external entities.

characteristics

intangible nature

- A software system has no physical existence: this contrasts with most systems from the other engineering branches.
- Only the functionality of software is perceptible.
- The software is not restricted by the materials properties.
- Software is developed, but not fabricated or constructed.
- Producing the first replica of a software system implies high costs, but the price goes down for the subsequent replicas.

no wear and tear

- Software does not wear out, since it does not lose its qualities.
- It exhibits an enormous degradation, derived from the changes that are made to keep it useful.
- The incorporation of new functionalities implies the introduction of defects in the software.

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Characterisation of the discipline

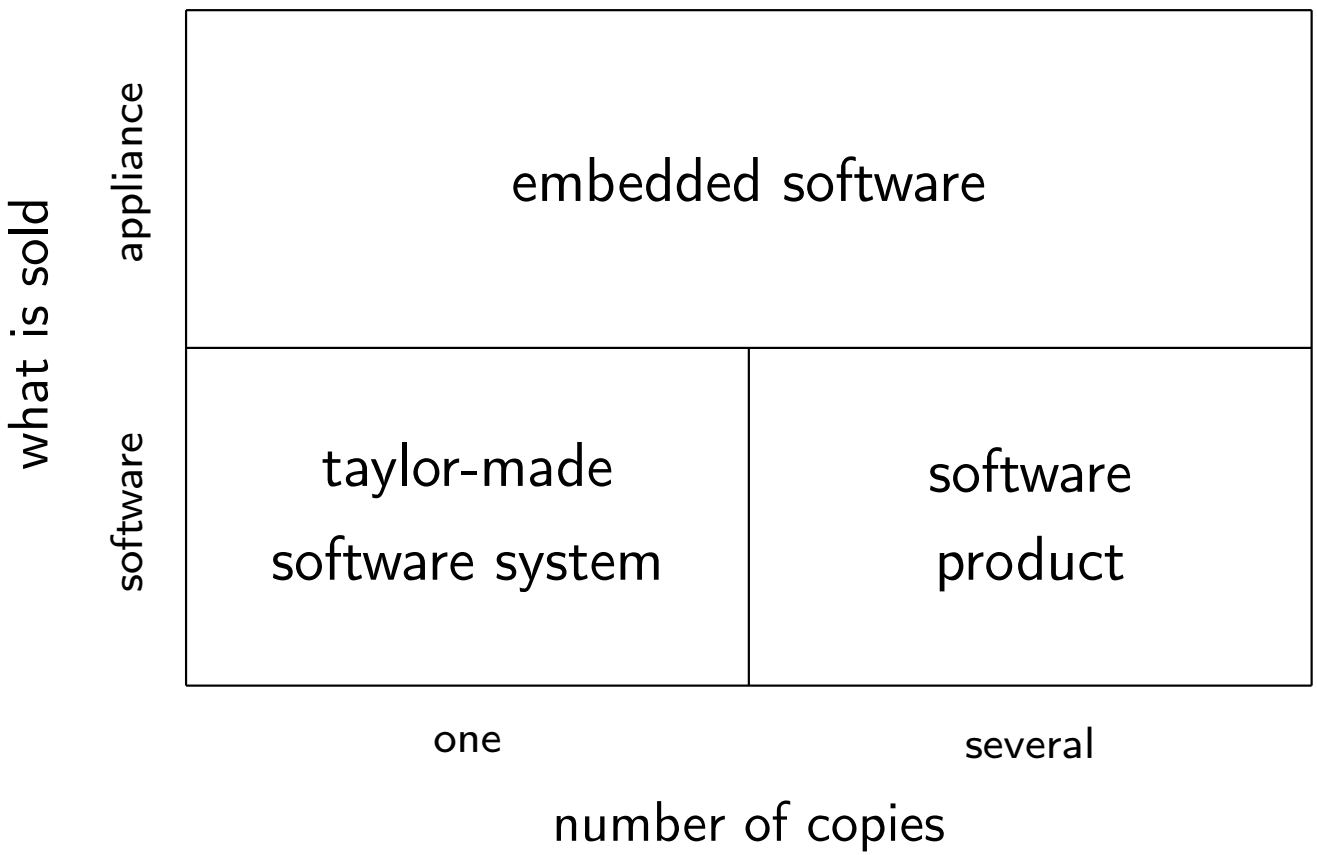
Software

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classification of software systems



classification of software systems







- The software included in any appliance is designated as embedded software.
- This term is used either for a unique appliance (satellite) or for devices produced in large numbers (television sets).
- These devices correspond to heterogeneous systems, rarely being referred to as 'software products', even in the cases where the software parts correspond to the most important technological dimension.
- Consumer electronics products include significant portions of software.
- The devaluation of the software technology results from the focus of the end users on the mechanical or electronic parts of those products.

classification of software systems

- Whenever the software system uses exclusively software technologies, then there are two types of systems.
- A tailor-made software system is developed by request of a given client for satisfying his own necessities.
- The principal objective is to satisfy the particular and specific needs of that client.
- A software product is produced to be commercialised for the public in general.
- A product is a combination of (material and immaterial) goods and services that the supplier combines, in accordance with his commercial interests, to transfer established rights to the client.

software product

A software product is composed of:

- **programs**, whose instructions when executed offer the functionalities of the product;
- **data structures** that permit programs to access the necessary information for their execution;
- **documentation** that explains how to install, use, and maintain the programs.

software product

- One can also classify software products in relation to the proximity that they have in relation to the hardware.
- The **system software** is responsible for managing the hardware resources of the computer.
- It is composed of programs that interact in an intense and direct way with the computer hardware.
- **Software applications** (apps) perform tasks that are useful for the end users.
- They support the realisation of the individual tasks of the persons and the execution of the organisational processes.
- These apps to be executed use a computer (hardware) and a system software.



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definition

- The area in which the system is explored is designated as domain.
- A domain can be considered as a business area, collection of problems, or knowledge area with its own terminology.

domain: area of human knowledge or activity that is characterised by possessing a set of concepts and terms that the respective players know.



- Before initiating the development of a system in a given domain, the requirements engineer may have no knowledge about that domain.

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problem and solution domains

- The problem domain is the context where one feels the necessities that need to be satisfied by the system to be developed.
- For a restaurant, the problem domain includes the elements that characterise it: clients, cooks, tables, chairs, towels, cutlery, crockery, meals, etc.



- The solution domain refers to the activities that are executed and the artefacts that are handled and constructed to solve the problem.

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classification of software systems domains

software systems domain are classified along five orthogonal axes:

1. difficulty of the problem class;
2. temporal relationship between data and processing;
3. number of tasks to be executed simultaneously;
4. relative difficulty of the data, interaction, and control aspects of the problem;
5. determinism level.

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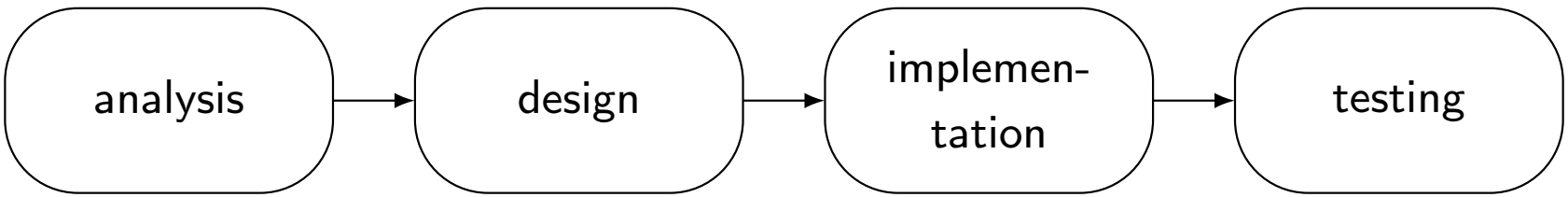
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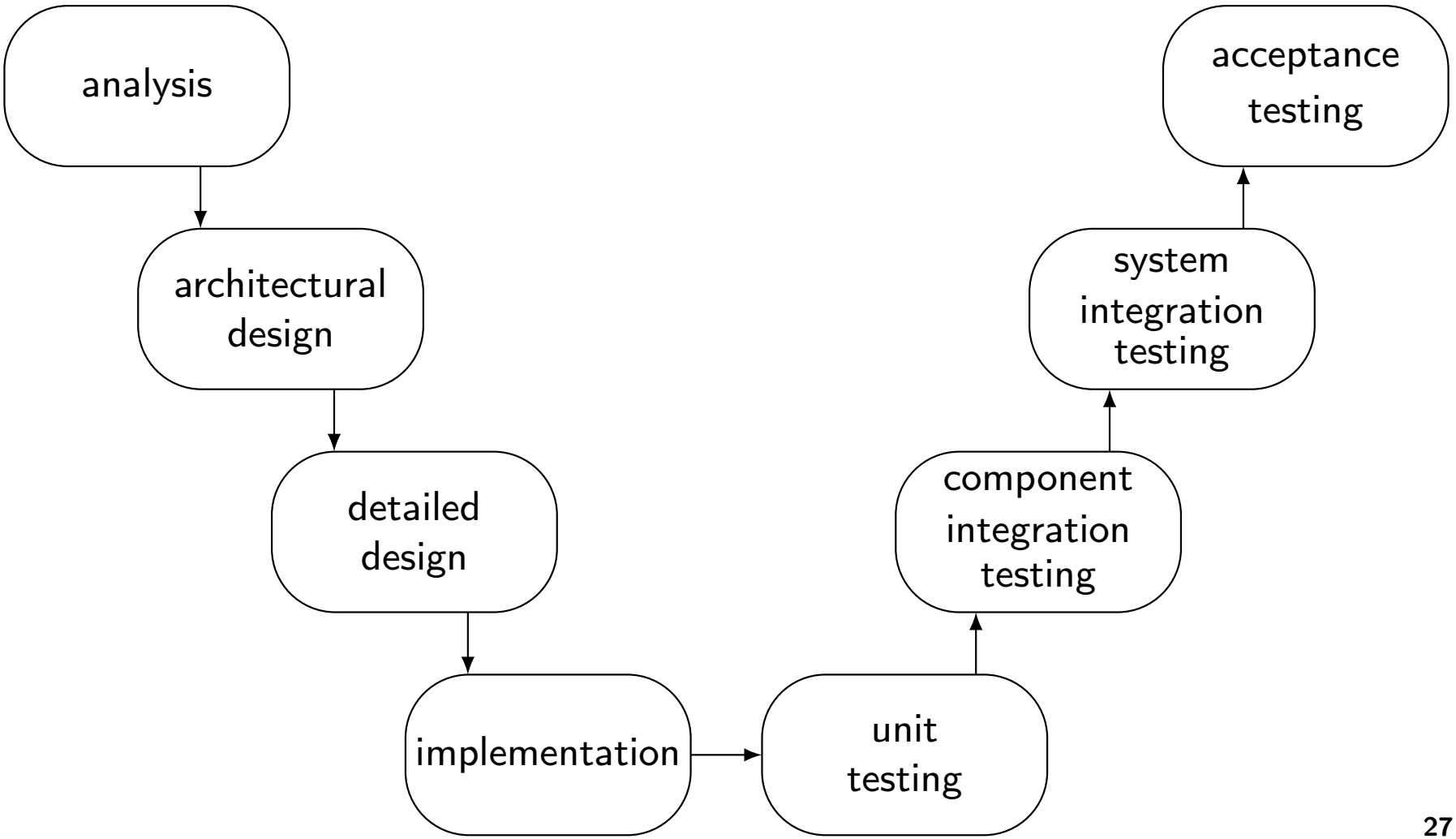
definition

- A *process model* represents a development process and indicates the way it must be organised.
- The process models aim to help the engineers in establishing the relation among the activities and the techniques that are part of the development process.
- Objectives:
  1. to clearly identify the activities that must be followed to develop a system;
  2. to introduce consistency in the development process, ensuring that the systems are developed with the same principles;
  3. to evaluate the obtained results and to verify the observance of the deadlines and the resources needs;
  4. to stimulate a bigger reuse of components, to increase the productivity of the development teams.

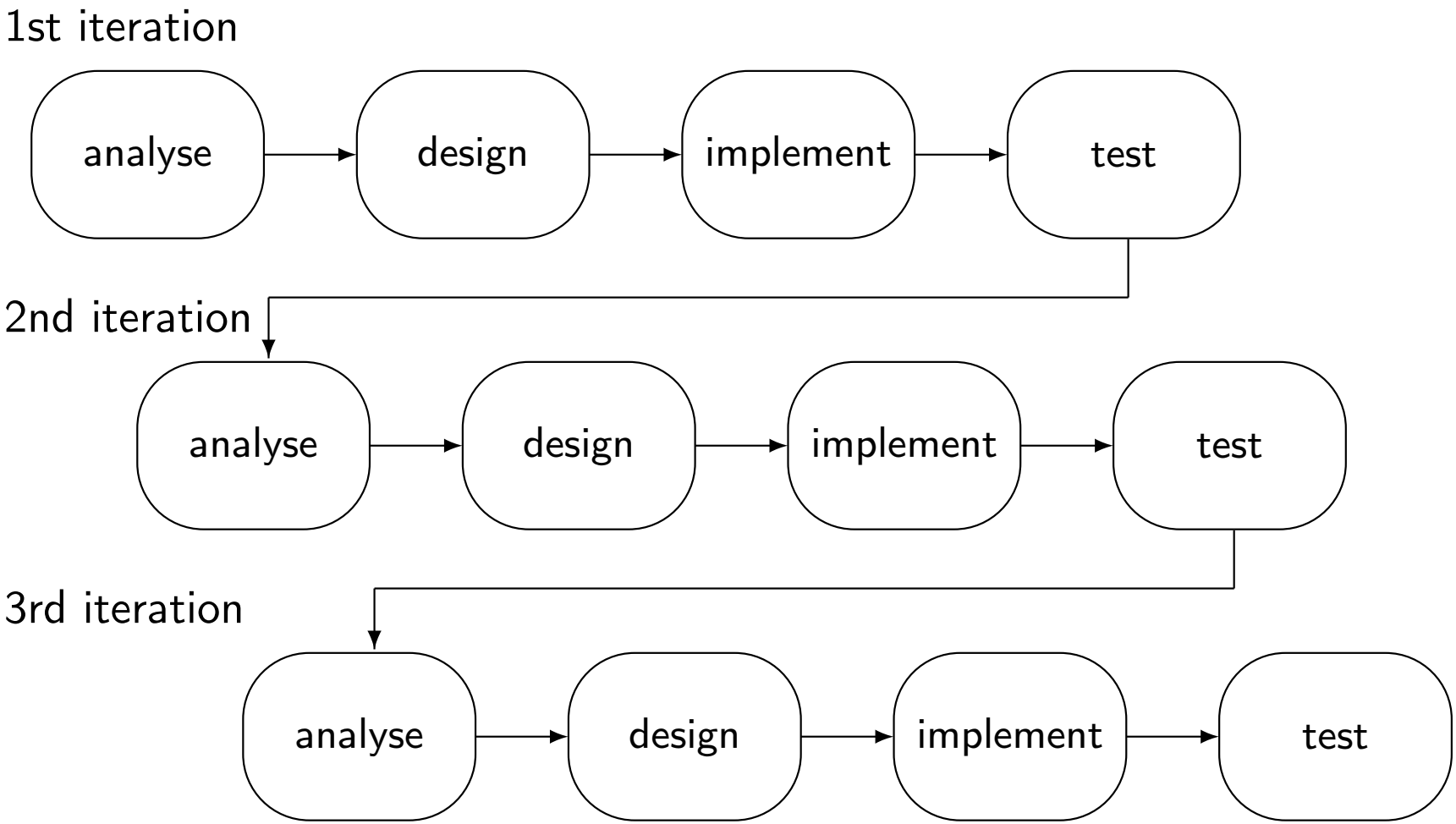
waterfall



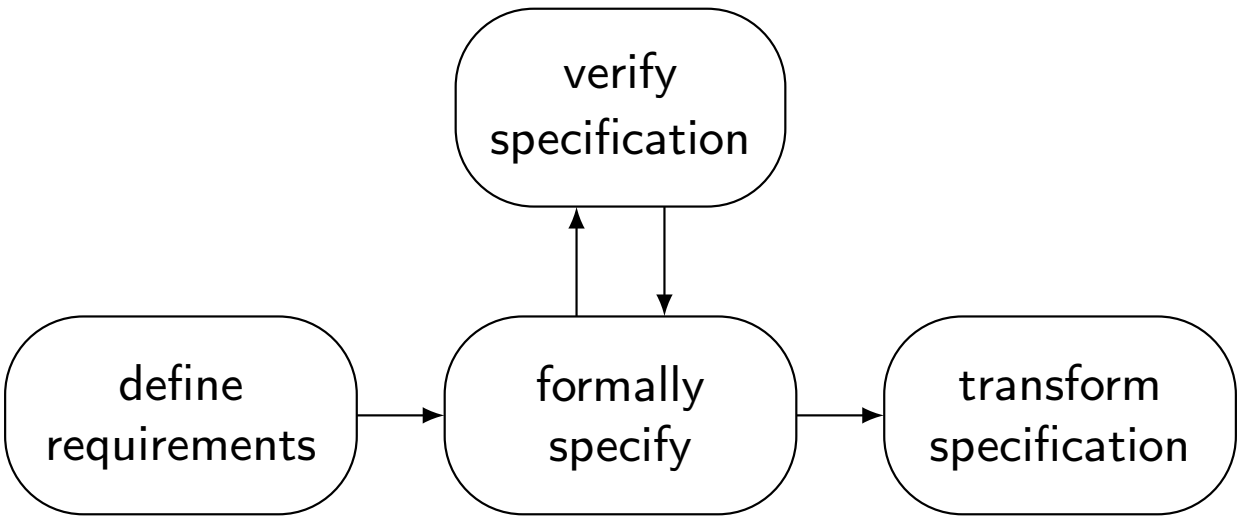
V model


analysis
architectural design
detailed design
implementation
unit testing
component integration testing
system integration testing
acceptance testing

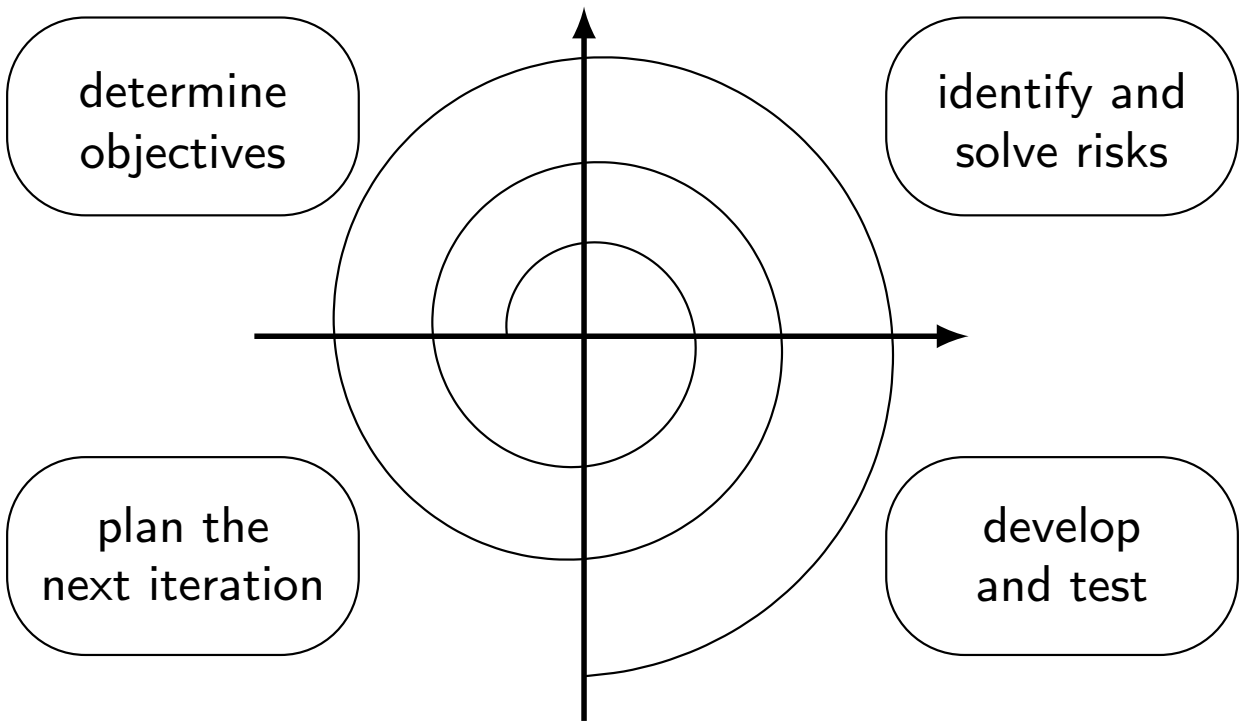
incremental and iterative



transformational



spiral


determine objectives
identify and solve risks
plan the next iteration
develop and test

Summary

- Software engineering is focused on all the aspects related to the development and production of software.
- Software engineering is not just centred in the technical aspects associated with software development, but includes also process management activities.
- Software engineering is a scientific area with an extremely important economic and social relevance and that has allowed people to solve many of their daily tasks.
- The software engineering body of knowledge is structured into 15 KAs according to the SWEBOK.
- Software is the principal artefact that results form the software engineering process.
- The software systems are developed to satisfy the stakeholders needs.
- The domain of a software system can be a business area, a collection of problems, or a knowledge area.

bibliography

- Fernandes JM and Machado RJ; *Requirements in engineering projects*, Springer, Lecture Notes in Management and Industrial Engineering series, ISBN 978-3-319-18596-5, 2016. [chapter 2]

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